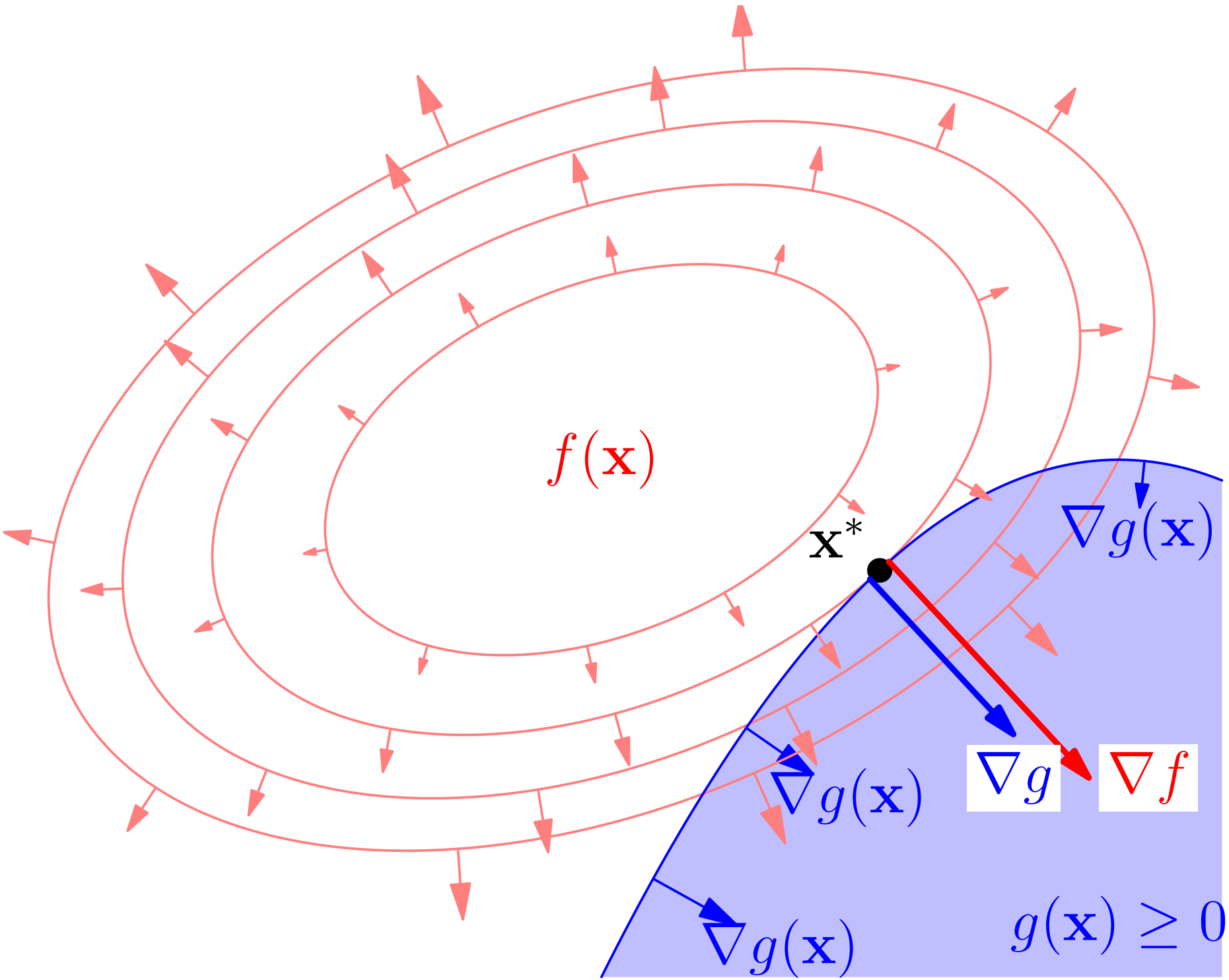$$\nabla f(\mathbf{x}^*) = \alpha \nabla g(\mathbf{x}^*),\ \alpha \approx 1.4 > 0$$